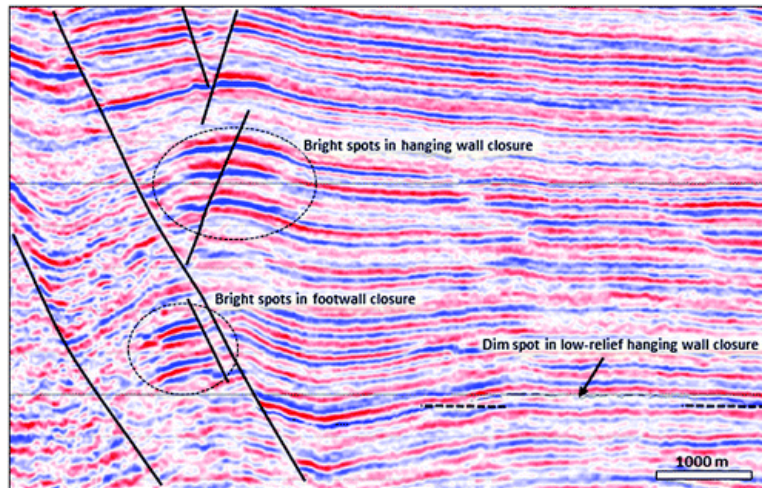




The Basics of Direct Hydrocarbon Indicators (DHIs)

Satyesh B.

La detección de petróleo es mi pasión



25 de julio de 2024

Satyesh Bhandari

Direct Hydrocarbon Indicators or DHIs, are specific features in seismic data that **suggest the presence of hydrocarbons** (oil and gas) in subsurface formations. These indicators help geophysicists identify potential hydrocarbon reservoirs without the immediate need for drilling. DHIs can significantly reduce exploration risk and cost.

Direct hydrocarbon indicators (DHIs) are seismic measurements that can indicate the presence of hydrocarbons in the subsurface. **They are anomalous seismic responses that occur when changes in pore fluids cause changes in the elastic properties of the rock.** DHIs **are most common in young, unconsolidated siliciclastic sediments** with large impedance across lithologic boundaries.

DHIs are analyzed in conjunction with other geophysical and geological data to help geoscientists prioritize drilling locations and identify targets. They can help inform decisions during the exploration and development of oil and gas reservoirs, but they are not conclusive evidence of hydrocarbons. **DHIs need to be further validated through additional data and analysis, such as well data, geological and geophysical data.**

Types of Direct Hydrocarbon Indicators

some common types of DHIs and their roles in seismic interpretation are listed here.

1. Amplitude Anomalies

- **Bright Spots:** Areas with **unusually relatively high seismic amplitude** that suggest the presence of hydrocarbons, often caused by **gas-filled sands**.

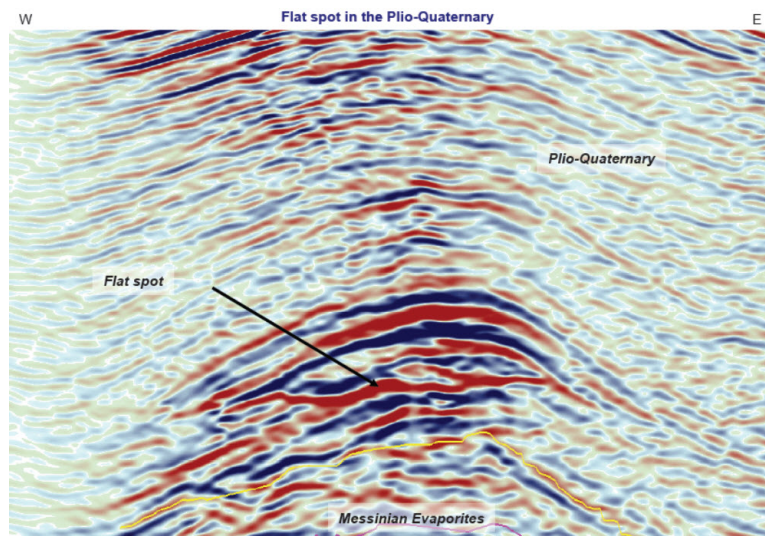
- **Dim Spots:** Areas with lower-than-expected amplitude, possibly indicating oil instead of gas or water saturation.
- **Flat Spots:** Horizontal reflections that can represent the interface between hydrocarbons and water within a reservoir.

2. Polarity Reversals

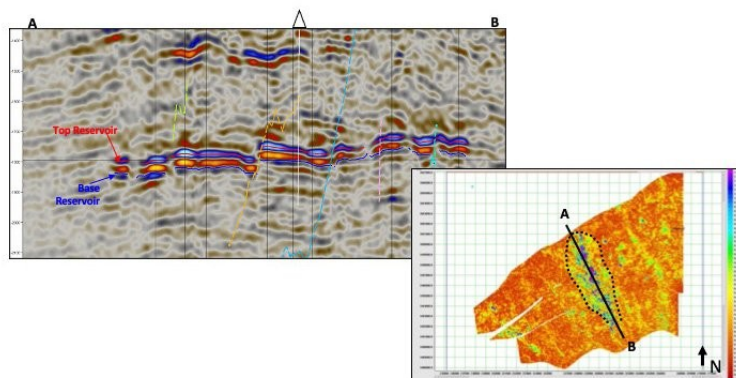
- Changes in the polarity of seismic reflections can indicate a change in the fluid content of the rock, such as a transition from water-bearing to hydrocarbon-bearing zones.

3. Frequency Anomalies

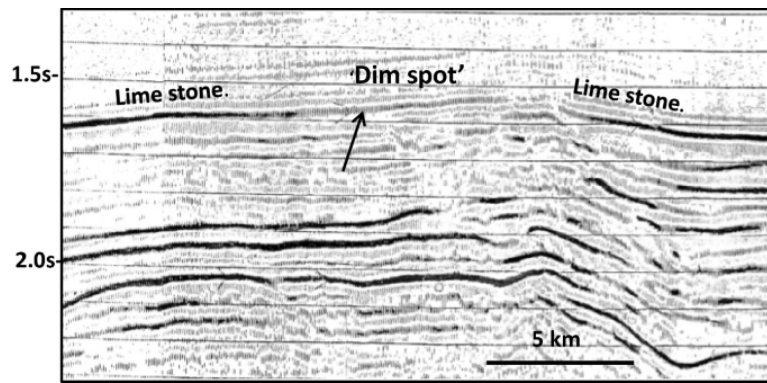
- **High-Frequency Shadows:** Zones where high-frequency seismic waves are absorbed or scattered, often associated with gas clouds above hydrocarbon reservoirs.
- **Low-Frequency Shadows:** Reduced frequencies below potential hydrocarbon zones, which can be indicative of gas or oil presence.



Amplitude Anomalies Identification and Extraction



Bright Spot indicating reservoir limit



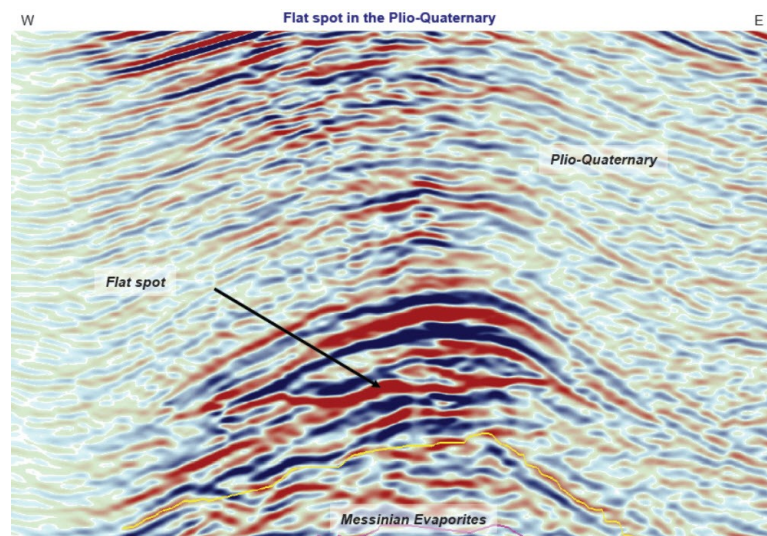
Dim Spot

4. Amplitude Versus Offset (AVO)

- Variations in reflection amplitude with changes in the angle or offset of seismic wave travel. Specific AVO responses can indicate the presence of hydrocarbons: **Class 1 AVO**: High impedance contrast, usually indicating gas sands. **Class 2 AVO**: Zero or low impedance contrast, suggesting oil or gas. **Class 3 AVO**: Strong decrease in amplitude with offset, typically indicating gas sands with higher porosity. **Class 4 AVO**: Increase in amplitude with offset, often pointing to gas-charged sands with a higher impedance than the surrounding rock.

5. Flat Events

- Reflectors that cut across structural features and are horizontal in nature, which can represent fluid contacts (gas-water, oil-water interfaces) within the reservoir.



Role of DHIs in Seismic Interpretation

1. Risk Reduction

- Identifying DHIs can help in reducing the risk associated with drilling by providing more evidence of hydrocarbon presence before drilling operations.

2. Reservoir Characterization

- DHIs contribute to a better understanding of the size, shape, and quality of the reservoir, helping in estimating resource, reserves and production potential.

3. Exploration Efficiency

- The presence of DHIs can work as exploration guide, focusing on more promising areas and thus saving time and resources.

4. Validation of Depositional Models

- DHIs provide critical data to validate and refine depositional and geophysical models, improving the overall accuracy of subsurface interpretations.

5. Decision Making

- The presence of strong DHIs can influence decisions on where to drill, how to design the well, and what production techniques to use.

Considerations and Limitations

- **False Positives:** Not all amplitude anomalies or DHIs necessarily indicate hydrocarbons; they can sometimes be caused by lithological changes, fractures, or other geological features. DHIs should not be taken as single guide for exploration drilling.
- **Integration with Other Data:** DHIs are most effective when used in conjunction with other geophysical, and petrophysical data.
- **Calibration:** Seismic data must be carefully calibrated with well data to accurately interpret DHIs.

In summary, DHIs are valuable tools in seismic interpretation, providing direct clues to the presence of hydrocarbons and aiding in the efficient and effective exploration and development of oil and gas resources.

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